Chapter 2 : Introduction

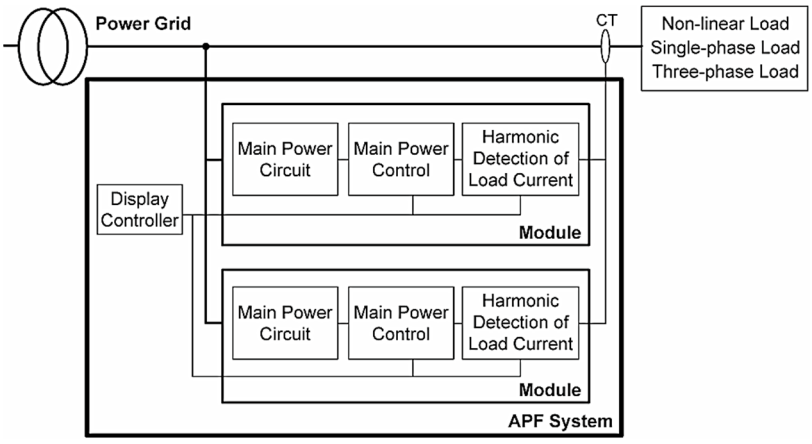
2.1 Product Introduction

The Delta PQC Series Mixed Fixed type APF, a harmonic compensator for the three-phase power grid, is applied to the harmonic control of the power grid. Features of the device are shown as below:

- Compensates 2^{nd} ~ 50^{th} harmonic currents (selectable).
- Compensates both inductive and capacitive reactive power.
- Correct three-phase load imbalance.
- Rapid dynamic responses, stable parameters and good harmonic compensation results.
- High efficiency and low loss. The advanced ECO mode allows you to save more energy.
- The modular design allows you to freely select different modules for the harmonic power compensation for different capacities.
- The system adopts an advanced 3-level structure and consists of digital signal processors (DSP), Field Programmable Gate Array (FPGA) and high power electronic devices, which has excellent performance and superior reliability.
- Supports remote on/off and fault monitoring via output relays and supports Modbus RTU/ TCP for remote SCADA monitoring.

Please see ***Figure 2-1*** for the APF system block diagram.

The APF system consists of multiple APF modules and one system cabinet. Load current is detected by external CT and load harmonic currents are calculated by APF controller. Based on the load current feedback, the controller manages power circuit to output inverse harmonic compensation current to cancel out load harmonic current. The system cabinet is equipped with a HMI Display that shows the information of individual modules and control the operation of individual modules. All power modules are paralleled connection and they are independently working.



(Figure 2-1: APF System Block Diagram)